

UNIT-IV Database, SQL and CSV Assessment

UNIT-IV: Database Concepts, SQL and CSV Files

Student Assessment Paper - Answers are not included.

Question type	Count	Marks each	Total
MCQ	30	1	30
Fill-ups	20	1	20
Simple coding	10	5	50
Total	60		100

Part A - Multiple Choice Questions

Question 1 - MCQ - 1 mark

A database is best described as:

- A. An organized collection of related data
- B. Only a single Python file
- C. A graphic design tool
- D. A temporary calculator memory

Question 2 - MCQ - 1 mark

Data is:

- A. Processed and meaningful facts
- B. Raw facts stored in a computer
- C. Only images stored in a file
- D. Only SQL commands

Question 3 - MCQ - 1 mark

Information is produced when data is:

- A. Deleted permanently
- B. Processed, organized and made meaningful
- C. Stored only in RAM
- D. Converted into binary operators

Question 4 - MCQ - 1 mark

DBMS stands for:

- A. Data Binary Management System
- B. Database Management System
- C. Digital Backup Machine Software
- D. Database Memory Server

Question 5 - MCQ - 1 mark

Which of the following is a common example of an RDBMS?

- A. MySQL
- B. MS Paint
- C. Notepad
- D. PowerPoint

Question 6 - MCQ - 1 mark

In relational database terminology, a table is also called a:

- A. Tuple
- B. Relation
- C. Attribute
- D. Key stroke

Question 7 - MCQ - 1 mark

In a table, a row is also called a:

- A. Tuple
- B. Attribute
- C. Field type
- D. Command

Question 8 - MCQ - 1 mark

In a table, a column is also called an:

- A. Tuple
- B. Attribute
- C. Record
- D. Database

Question 9 - MCQ - 1 mark

Which component of DBMS refers to physical devices like disk and I/O channels?

- A. Software
- B. Hardware
- C. Procedures
- D. Data Access Language

Question 10 - MCQ - 1 mark

Which data model represents data as a tree-like parent-child structure?

- A. Relational model
- B. Hierarchical model
- C. Network model
- D. Object model

Question 11 - MCQ - 1 mark

SQL stands for:

- A. Simple Query List
- B. Structured Query Language
- C. System Query Logic
- D. Sequential Query Link

Question 12 - MCQ - 1 mark

SQL is mainly used to:

- A. Edit photographs
- B. Access and manipulate databases
- C. Play audio files
- D. Compress videos

Question 13 - MCQ - 1 mark

Which SQL command is used to create a database?

- A. MAKE DATABASE
- B. CREATE DATABASE
- C. NEW DB
- D. BUILD TABLE

Question 14 - MCQ - 1 mark

Which SQL command is used to select a database for use?

- A. OPEN
- B. USE
- C. CHOOSE
- D. LOAD

Question 15 - MCQ - 1 mark

DDL stands for:

- A. Data Definition Language
- B. Data Display Language
- C. Database Division Logic
- D. Direct Data Link

Question 16 - MCQ - 1 mark

Which command belongs to DDL?

- A. CREATE
- B. INSERT
- C. UPDATE
- D. SELECT

Question 17 - MCQ - 1 mark

Which command belongs to DML?

- A. GRANT
- B. INSERT
- C. COMMIT
- D. CREATE

Question 18 - MCQ - 1 mark

Which SQL command displays records from a table?

- A. SELECT
- B. DROP
- C. ROLLBACK
- D. GRANT

Question 19 - MCQ - 1 mark

Which command is used to withdraw permissions granted to a user?

- A. COMMIT
- B. REVOKE
- C. SAVEPOINT
- D. TRUNCATE

Question 20 - MCQ - 1 mark

Which TCL command saves a transaction permanently?

- A. ROLLBACK
- B. COMMIT
- C. GRANT
- D. ALTER

Question 21 - MCQ - 1 mark

CSV stands for:

- A. Comma Separated Values
- B. Computer Saved Variables
- C. Column System View
- D. Common Storage Version

Question 22 - MCQ - 1 mark

A CSV file is usually:

- A. A plain text file
- B. A binary image file
- C. A compiled Python file
- D. A database server

Question 23 - MCQ - 1 mark

In a normal CSV file, fields are commonly separated by:

- A. Commas
- B. Semicolons only
- C. Tabs only
- D. Full stops only

Question 24 - MCQ - 1 mark

If a CSV field contains a comma, the field should be:

- A. Deleted
- B. Enclosed in double quotes
- C. Written in square brackets
- D. Converted into SQL

Question 25 - MCQ - 1 mark

Which Python module is used to work with CSV files?

- A. csv
- B. math
- C. random
- D. ospath

Question 26 - MCQ - 1 mark

Which object is commonly used to read rows from a CSV file?

- A. csv.reader
- B. csv.timer
- C. csv.scan
- D. csv.openbook

Question 27 - MCQ - 1 mark

Which object is commonly used to write rows to a CSV file?

- A. csv.writer
- B. csv.printer
- C. csv.saver
- D. csv.table

Question 28 - MCQ - 1 mark

CSV files cannot store:

- A. Plain text data
- B. Rows and columns
- C. Charts, formulas and formatting
- D. Comma-separated values

Question 29 - MCQ - 1 mark

If double quotes are part of data inside a quoted CSV field, they must be:

- A. Removed always
- B. Doubled
- C. Converted into spaces
- D. Written as brackets

Question 30 - MCQ - 1 mark

In CSV terminology, each line is usually treated as a:

- A. Row/record
- B. Database engine
- C. Python package
- D. SQL privilege

Part B - Fill in the Blanks

Question 31 - FILLUP - 1 mark

A database is an organized collection of related _____.

Answer: _____

Question 32 - FILLUP - 1 mark

Raw facts stored in a computer are called _____.

Answer: _____

Question 33 - FILLUP - 1 mark

Processed data that gives meaning is called _____.

Answer: _____

Question 34 - FILLUP - 1 mark

DBMS stands for Database _____ System.

Answer: _____

Question 35 - FILLUP - 1 mark

RDBMS stores data mainly in the form of _____.

Answer: _____

Question 36 - FILLUP - 1 mark

A table in relational database is known as a _____.

Answer: _____

Question 37 - FILLUP - 1 mark

A row in a table is known as a _____.

Answer: _____

Question 38 - FILLUP - 1 mark

A column in a table is known as an _____.

Answer: _____

Question 39 - FILLUP - 1 mark

The _____ data model represents one-to-many parent-child relationships.

Answer: _____

Question 40 - FILLUP - 1 mark

The process of reducing unnecessary repetition of data is called _____.

Answer: _____

Question 41 - FILLUP - 1 mark

SQL stands for Structured Query _____.

Answer: _____

Question 42 - FILLUP - 1 mark

The SQL command used to create a database is _____.

Answer: _____

Question 43 - FILLUP - 1 mark

The SQL command used to choose a database is _____.

Answer: _____

Question 44 - FILLUP - 1 mark

DDL stands for Data _____ Language.

Answer: _____

Question 45 - FILLUP - 1 mark

DML stands for Data _____ Language.

Answer: _____

Question 46 - FILLUP - 1 mark

The DQL command used to display records is _____.

Answer: _____

Question 47 - FILLUP - 1 mark

The DCL command that gives permission is _____.

Answer: _____

Question 48 - FILLUP - 1 mark

The DCL command that withdraws permission is _____.

Answer: _____

Question 49 - FILLUP - 1 mark

The TCL command that permanently saves a transaction is _____.

Answer: _____

Question 50 - FILLUP - 1 mark

CSV stands for Comma Separated _____.

Answer: _____

Part C - Simple Coding Questions

Question 51 - CODING - 5 marks

Write a function `make_record(stuno, name, age)` that returns a dictionary with keys 'StuNo', 'StuName', and 'StuAge'.

Starter code

```
def make_record(stuno, name, age):  
    # write your code here  
    pass
```

Visible test cases

```
Input: make_record(1101, "Kannan", 14)
Expected: {'StuNo': 1101, 'StuName': 'Kannan', 'StuAge': 14}

Input: make_record(1103, "Vidhya", 15)
Expected: {'StuNo': 1103, 'StuName': 'Vidhya', 'StuAge': 15}
```

Expected output / task behavior

Return a dictionary representing one table row/record.

Question 52 - CODING - 5 marks

Write a function `table_shape(rows)` that accepts a list of dictionaries and returns a tuple (`number_of_rows`, `number_of_columns`). Return (0, 0) for an empty list.

Starter code

```
def table_shape(rows):
    # write your code here
    pass
```

Visible test cases

```
Input: table_shape([{'id':1, 'name':'A'}, {'id':2, 'name':'B'}])
Expected: (2, 2)

Input: table_shape([])
Expected: (0, 0)
```

Expected output / task behavior

Return row count and column count.

Question 53 - CODING - 5 marks

Write a function `create_database_sql(name)` that returns the SQL command to create a database with the given name.

Starter code

```
def create_database_sql(name):
    # write your code here
    pass
```

Visible test cases

```
Input: create_database_sql("stud")
Expected: CREATE DATABASE stud;

Input: create_database_sql("school")
Expected: CREATE DATABASE school;
```

Expected output / task behavior

Return `CREATE DATABASE <name>;`

Question 54 - CODING - 5 marks

Write a function `use_database_sql(name)` that returns the SQL command to use/select the given database.

Starter code

```
def use_database_sql(name):
    # write your code here
    pass
```

Visible test cases

```
Input: use_database_sql("stud")
Expected: USE stud;

Input: use_database_sql("library")
Expected: USE library;
```

Expected output / task behavior

Return `USE <name>;`

Question 55 - CODING - 5 marks

Write a function `select_all_sql(table_name)` that returns the SQL command to display all records from a table.

Starter code

```
def select_all_sql(table_name):
    # write your code here
    pass
```

Visible test cases

```
Input: select_all_sql("student")
Expected: SELECT * FROM student;
```

```
Input: select_all_sql("marks")
Expected: SELECT * FROM marks;
```

Expected output / task behavior

Return SELECT * FROM <table_name>;

Question 56 - CODING - 5 marks

Write a function insert_student_sql(stuno, name, age) that returns an INSERT command for table student with fields StuNo, StuName, and StuAge.

Starter code

```
def insert_student_sql(stuno, name, age):
    # write your code here
    pass
```

Visible test cases

```
Input: insert_student_sql(1101, "Kannan", 14)
Expected: INSERT INTO student (StuNo, StuName, StuAge) VALUES (1101, 'Kannan', 14);

Input: insert_student_sql(1102, "Ram", 15)
Expected: INSERT INTO student (StuNo, StuName, StuAge) VALUES (1102, 'Ram', 15);
```

Expected output / task behavior

Return a correctly formatted INSERT command.

Question 57 - CODING - 5 marks

Write a function parse_csv_lines(text) that accepts CSV text and returns a list of rows using the csv module.

Starter code

```
def parse_csv_lines(text):
    import csv
    import io
    # write your code here
    pass
```

Visible test cases

```
Input: parse_csv_lines("Name, Age\nKannan, 14")
Expected: [['Name', 'Age'], ['Kannan', '14']]

Input: parse_csv_lines("A,B\n1,2")
Expected: [['A', 'B'], ['1', '2']]
```

Expected output / task behavior

Return CSV rows as a list of lists.

Question 58 - CODING - 5 marks

Write a function count_csv_records(text) that returns the number of data records in CSV text, excluding the header row.

Starter code

```
def count_csv_records(text):
    import csv
    import io
    # write your code here
    pass
```

Visible test cases

```
Input: count_csv_records("RollNo, Name\n12101, Nivetha\n12102, Lavanya")
Expected: 2

Input: count_csv_records("A,B\n1,2")
Expected: 1
```

Expected output / task behavior

Return number of rows after the header.

Question 59 - CODING - 5 marks

Write a function to_csv_row(values) that converts a list of values into one CSV row string using csv.writer.

Starter code

```
def to_csv_row(values):
    import csv
    import io
    # write your code here
    pass
```

Visible test cases

```
Input: to_csv_row(["Ram", "Gopalapuram, Chennai"])
Expected: Ram,"Gopalapuram, Chennai"
```

```
Input: to_csv_row(["A", "B"])
Expected: A,B
```

Expected output / task behavior

Return a single CSV-formatted row without the ending newline.

Question 60 - CODING - 5 marks

Write a function `csv_to_dicts(text)` that reads CSV text with a header and returns a list of dictionaries.

Starter code

```
def csv_to_dicts(text):
    import csv
    import io
    # write your code here
    pass
```

Visible test cases

```
Input: csv_to_dicts("Name,Age\nKannan,14")
Expected: [{'Name': 'Kannan', 'Age': '14'}]
```

```
Input: csv_to_dicts("RollNo,Name\n12101,Nivetha")
Expected: [{'RollNo': '12101', 'Name': 'Nivetha'}]
```

Expected output / task behavior

Return `list(csv.DictReader(...))`.